

Data Analyst Capstone Project Report

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OUTLINE



- Executive Summary
- Introduction
- Methodology
- Results
 - Visualization – Charts
 - Dashboard
- Discussion
 - Findings & Implications
- Conclusion
- Appendix

EXECUTIVE SUMMARY



- Data contextualization and analysis goal
- Methodology description
 - Data gathering
 - Data analysis
 - Data visualization
- Results shows the graphs and trend supported by various graphs
- Discussion of overall findings and implications regarding the results previously exposed
- Final conclusions of the findings in Dataset.
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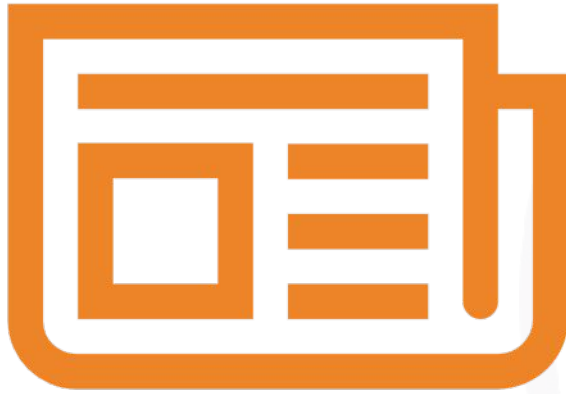
INTRODUCTION



Stack Overflow's annual Developer Survey is the largest and most comprehensive survey of people who code around the world

- Result don't represent everyone in the developer community evenly
- Nearly 90,000 developers responded
- Trends
 - Technology worked with
 - Future technology to work

METHODOLOGY



- Collect survey data
 - Web scraping
 - APIs
 - Request library
- Data wrangling
- Exploratory Data Analysis (EDA)
 - Distribution
 - Outlier Handling
 - Correlation
- Visualization
 - Distribution of data
 - Relationship
 - Comparison
 - Composition
- Dashboards



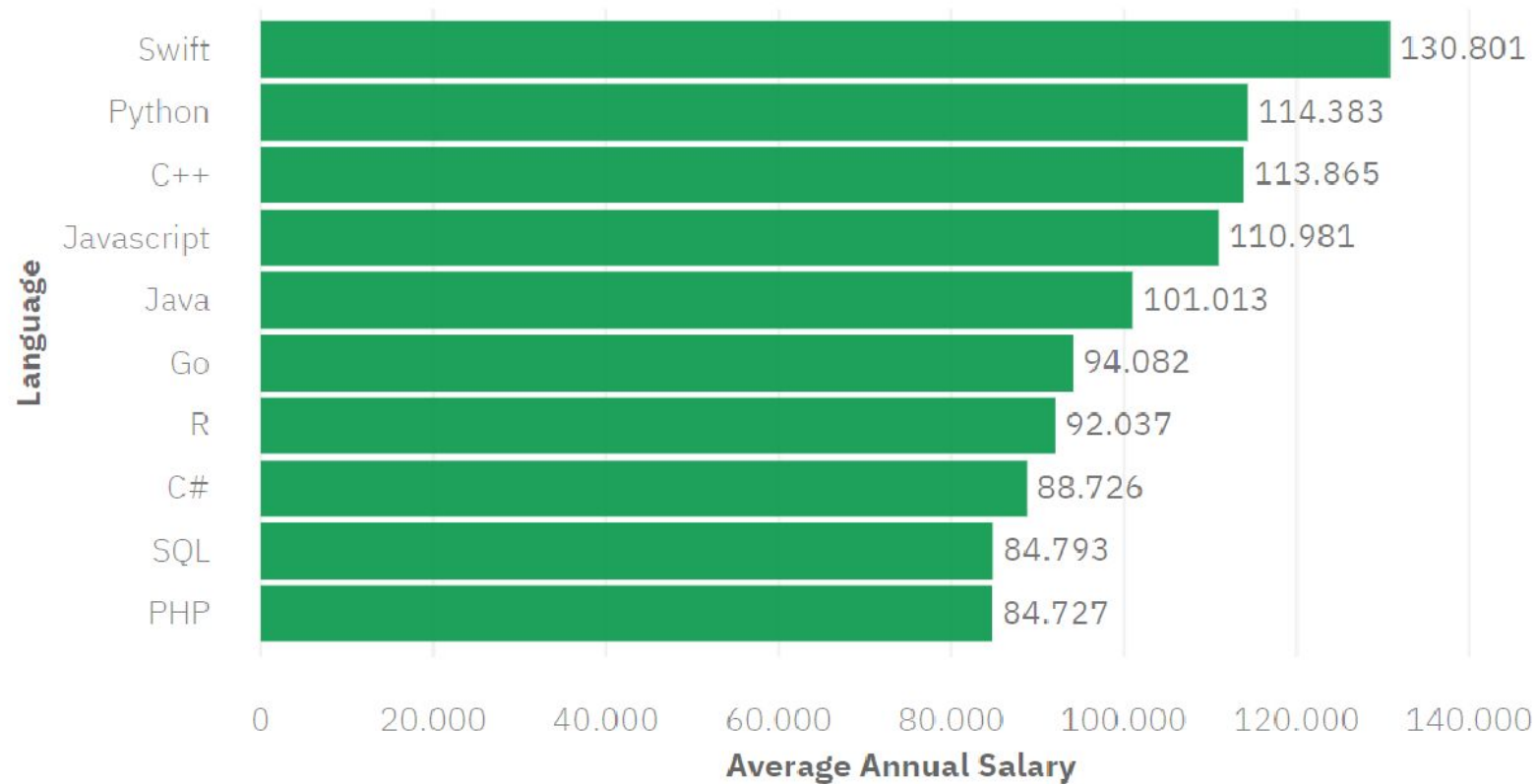
Results



JOB POSTINGS

In Module 1 you have collected the job posting data using Job API in a file named “job-postings.xlsx”. Present that data using a bar chart here. Order the bar chart in the descending order of the number of job postings.

POPULAR LANGUAGES AND AVG SALARY

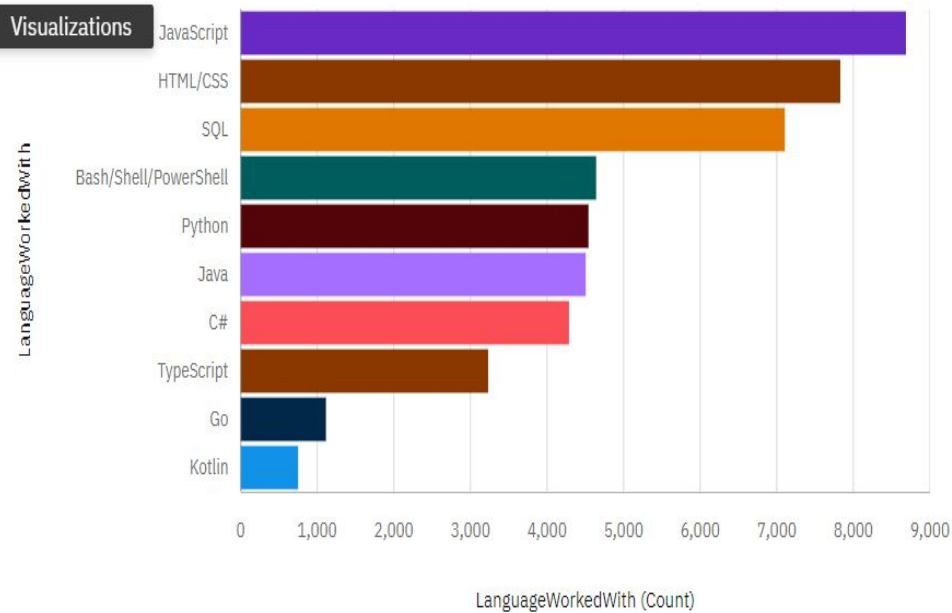


PROGRAMMING LANGUAGE TRENDS - CURRENT AND NEXT YEAR

Summarize key trends shown in the charts.

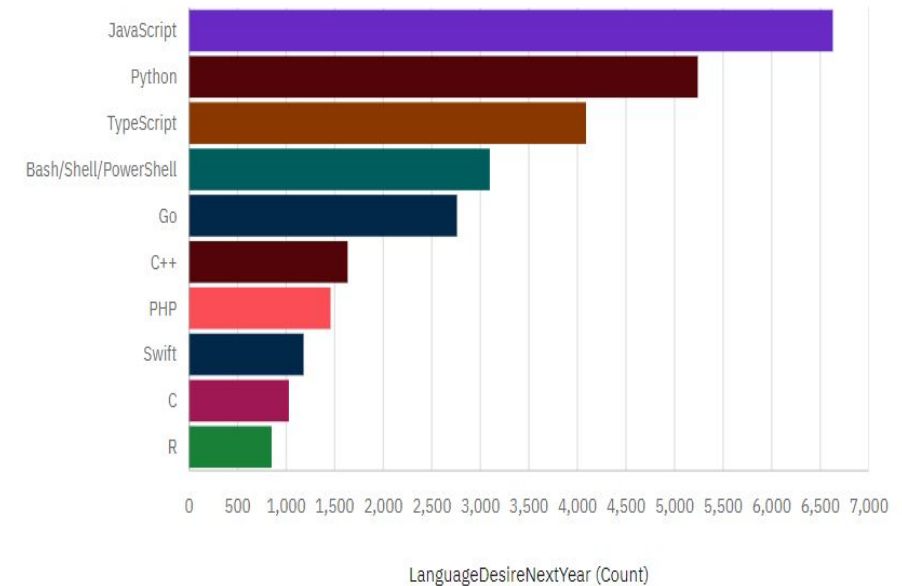
Current Year

Top 10 Language Worked with



Next Year

Top 10 Language Desire Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- Persistain interest in Javascript
- Decline in C++
- Rise in Python

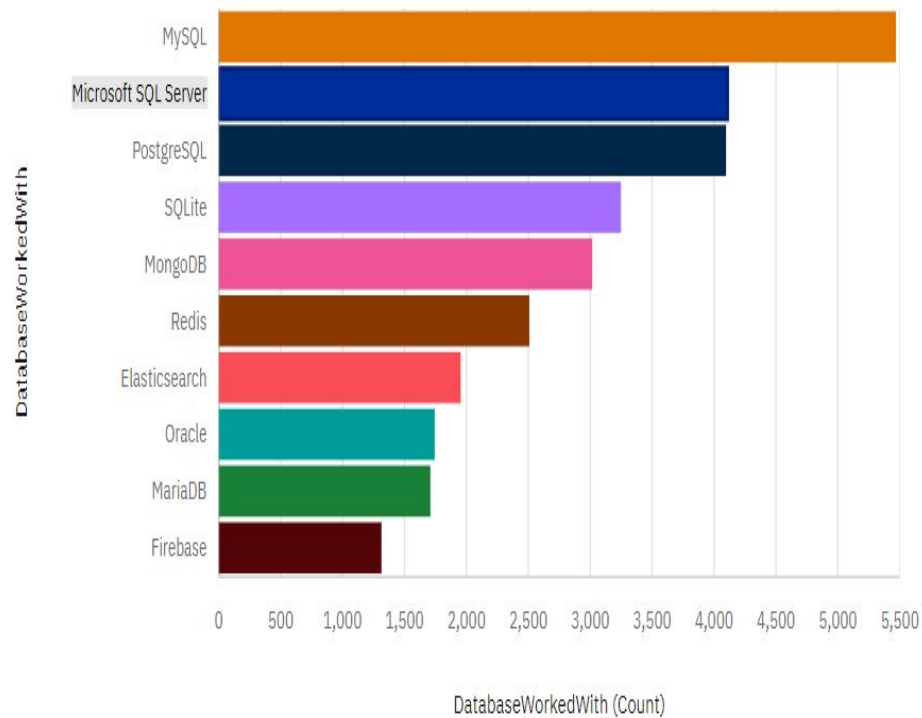
Implications

- Javascript is going to continue to be utilized
- C++ is declining and programmers should focus on other languages
- Python is going to become a rising demand for employers

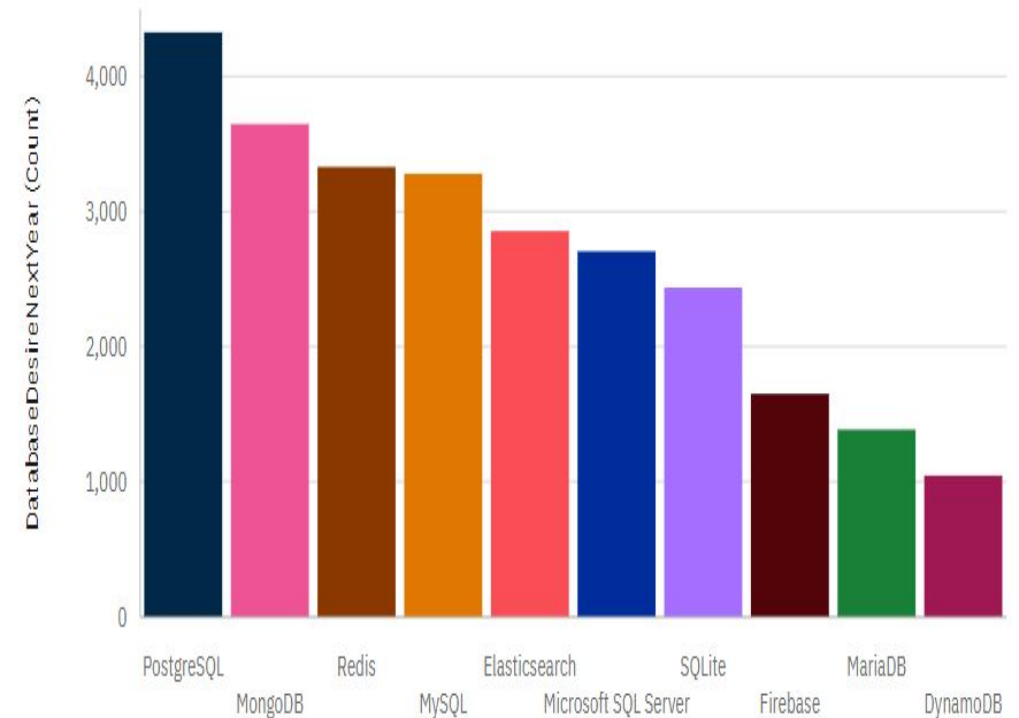
DATABASE TRENDS- CURRENT AND NEXT YEAR

Summarize key trends shown in the charts.

Current Year



Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- Rise in PostgreSQL
- Decline in MicrosoftSQL Server
- Mongo is rising at a high rate

Implications

- Postgres is a desired database
- MicrosoftSQL is becoming less popular among employers
- Mongo may become a top contender in the future

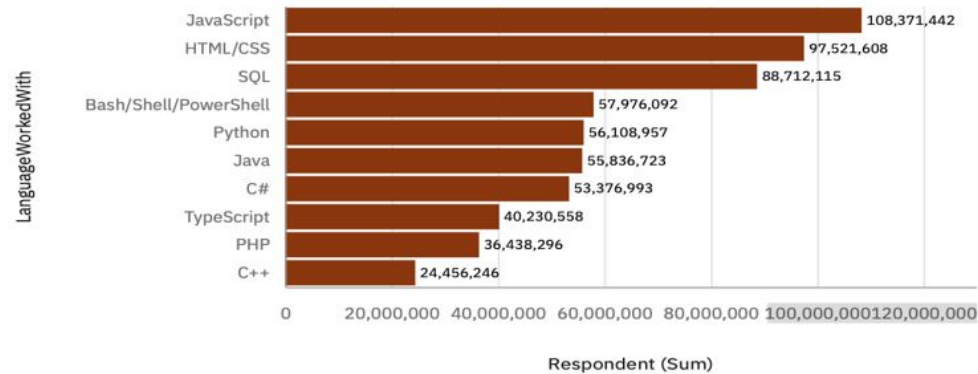
DASHBOARD



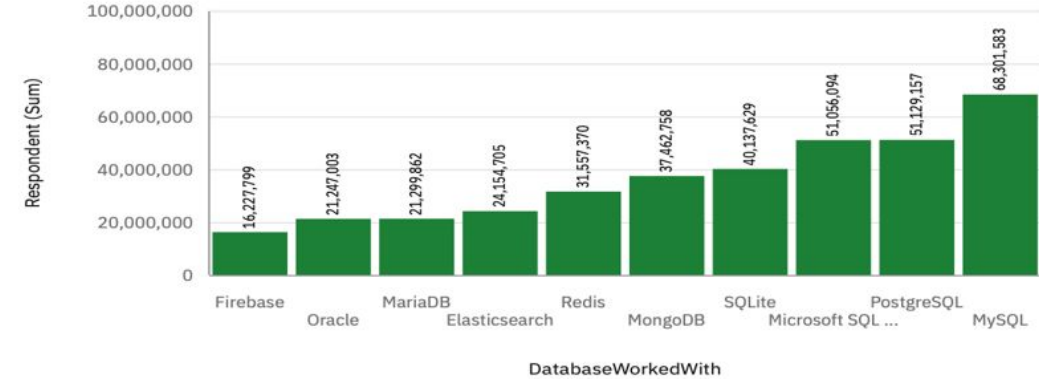
<Please present your dashboard in the following slides.>

DASHBOARD TAB 1: Current Technology Usage

Top 10 Languages Used by Respondents



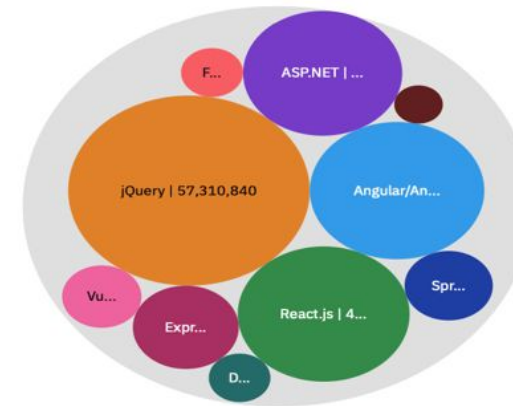
Top 10 Database Used by Respondents



Platforms Used by Respondents

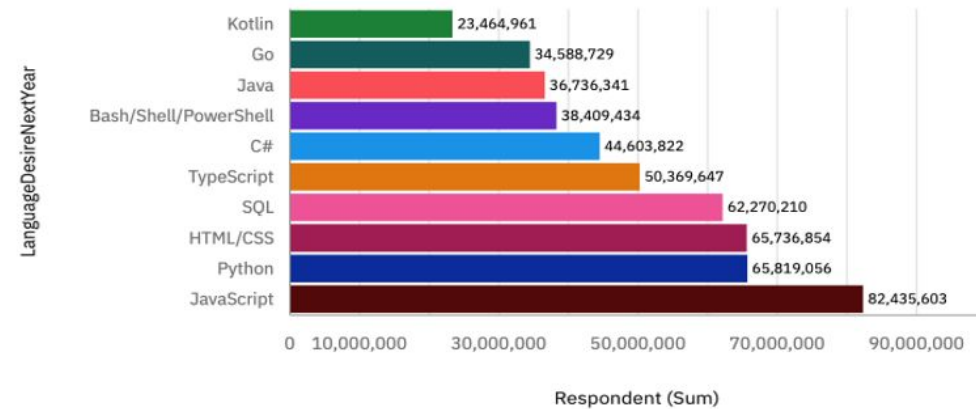


Top 10 Web Frames Used

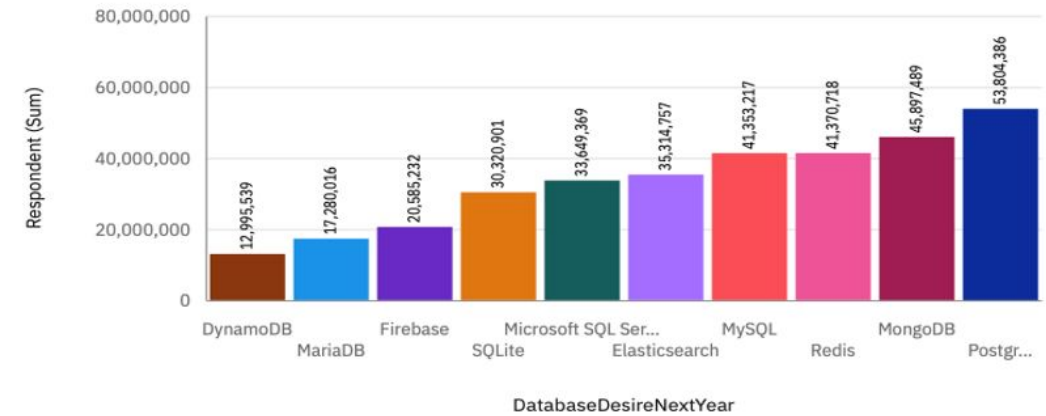


DASHBOARD TAB 2: Future Technology Trends

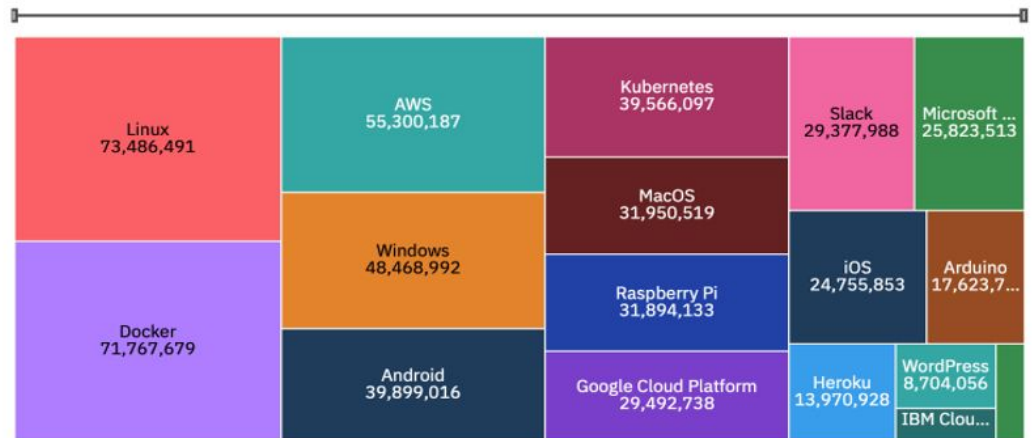
Top 10 Languages Respondents Want to Use in the Following Year



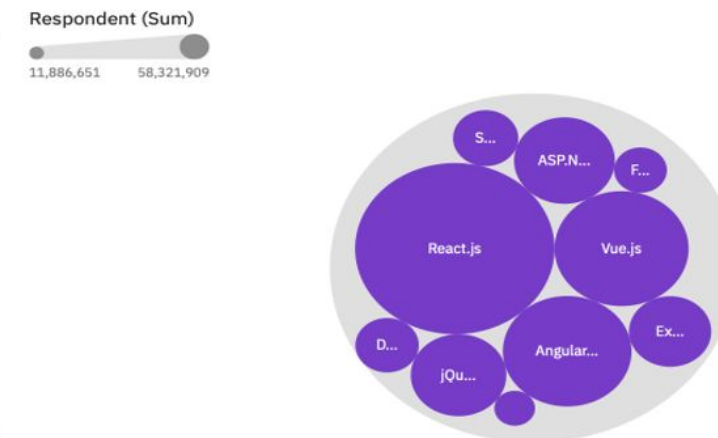
Top 10 Databases Respondents Want to Use in the Following Year



Platforms Respondents Want to Use in the Following Year



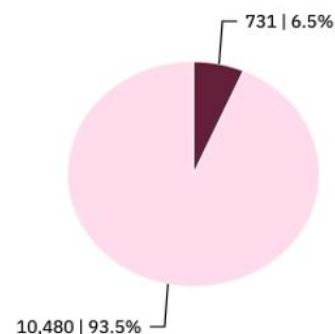
To 10 Web Frames Respondents Want to Use in the Following Year



DASHBOARD TAB 3: Demographics

Gender of Respondents

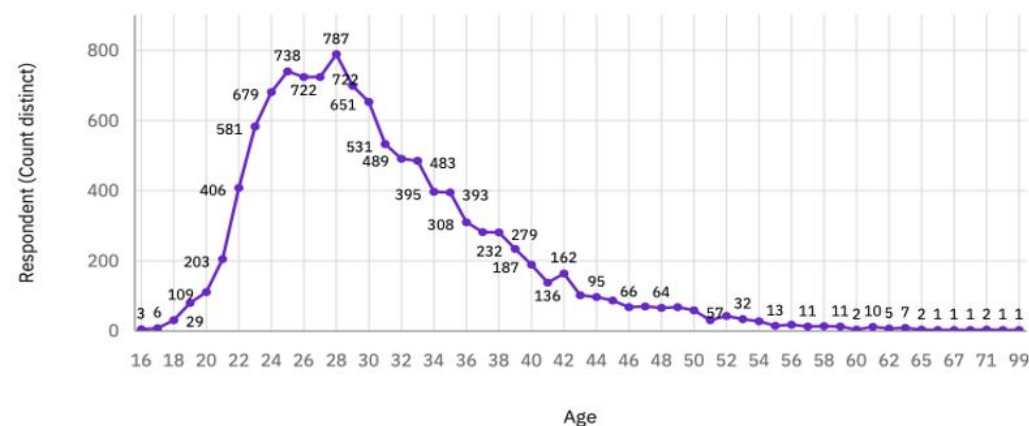
Gender
 ● Woman ● Man



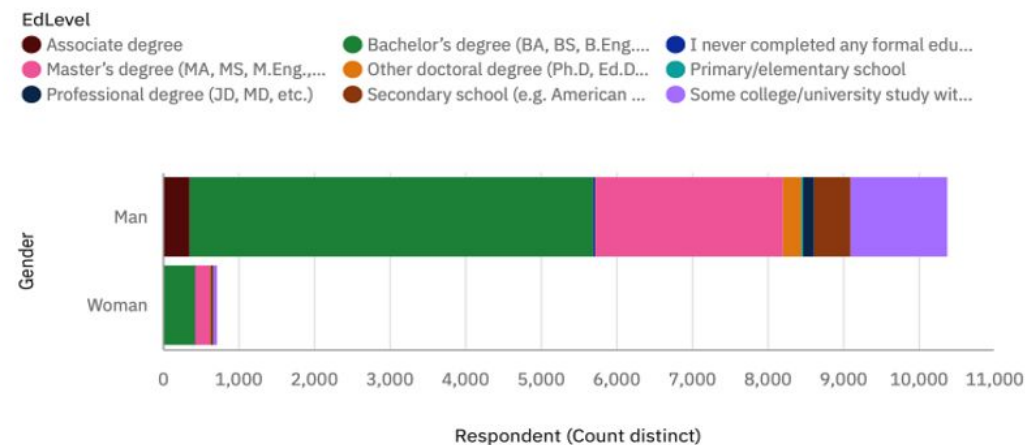
Countries of Respondents



Age Distribution of Respondents



Gender and Education Level of Respondents



DISCUSSION QUESTIONS



- What are the current trends in technology?
- What are the future trends in technology?
- What are the demographics for technology?

OVERALL FINDINGS & IMPLICATIONS

Findings

- Most popular languages are JavaScript, HTML, and SQL
- There is a rising interest in Kotlin, Go, and TypeScript
- The gender distribution is skewed towards men

Implications

- IWeb development continues to be the most used tools
- There is a rise in skills with mobile development
- There is a limitation in diversity and perspectives

CONCLUSION



- **Web Development and Cloud Domination:** JavaScript, HTML/CSS, and SQL are extensively used, indicating web development's stronghold, while platforms like AWS, Linux, and Docker suggest the growing prominence of cloud computing and containerization in modern tech stacks.
- **Emerging Tech Skills and Database Preferences:** Kotlin, Go, and TypeScript are gaining traction, signalling the need for up-skilling in mobile and systems development. Open-source databases like MySQL and PostgreSQL are top choices, highlighting their scalability and flexibility.
- **Demographic Imbalance and Opportunities:** The significant gender gap (93.5% men) and younger workforce (25-30 age peak) in tech underline the need for more diversity and inclusion efforts, offering potential for innovation through varied perspectives.